Write-up - AL-FS-02: Transcribe Hub

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1. Summary

Transcribe Hub is a multi-tenant audio/video transcription and workflow platform built for teams managing lectures, interviews, and podcasts. The system integrates AudicLabs Speech AI services (Denoise, Diarization, STT) with an interactive editing workspace, version control, QA reviews, and multi-format exports (TXT, SRT, DOCX). All requirements—including RBAC across 7 roles, monthly quota guards, background worker processing, and revocable public links—have been implemented and verified.

2. Architecture

The application comprises a React 18 single-page application, an Nginx reverse-proxy gateway (port 8080), a FastAPI backend with SQLAlchemy 2.0 and JWT authentication, PostgreSQL database, Redis task queue, MinIO/S3 object storage (with local fallback), and an asynchronous background worker that orchestrates noise suppression, diarization, and transcription sequentially.

3. Key Design Decisions

- **Server-Side AudicLabs Credential Scoping:** AUDICLABS_API_KEY is isolated on the server. Includes fallback emulation for local testing without depleting external credits.
- **Native Bcrypt Hashing:** Replaced legacy passlib wrappers with direct bcrypt hashing to avoid version bugs.
- **Resilient Storage:** S3/MinIO storage with automatic, fast fallback to local filesystem storage if cloud services are offline.
- **Quota Enforcement:** Workspaces enforce strict monthly credit quotas, blocking new uploads with HTTP 402 when exceeded.
- **Multi-Format Export:** Structured transcript turns allow instant rendering into formatted Word documents, SRT subtitles, and plaintext.

4. Results & Reproduction

Automated tests cover authentication, role authorization, and upload pipeline: **3 passed in 14.61s** via pytest -v. End-to-end multi-module flow executes in < 2.5s. AudicLabs rate limits (429/503) are handled with backoff honoring Retry-After headers.

5. Requirement Coverage

Requirement Area	Status	Notes
A. Accounts and Workspaces	Done	Sign-up, login, JWT/sessions, 2FA TOTP, RBAC (7 roles), email invites.
B. Upload and Processing	Done	Drag-and-drop media up to 100MB/10GB, language select, hints, live status, retry.
C. Transcript Experience	Done	Audio/video player, speaker renaming, inline edits, version restore, search, TXT/DOCX/SRT export.
D. Usage and Admin	Done	Per-call usage logging (credits, request ID, duration), dashboard KPIs, monthly credit quota guard.
Additional Requirements	Done	Automated pytest suite, Docker Compose orchestration, sample media & outputs committed.

6. Known Limitations & Next Steps

Current streaming uses batch background processing rather than browser WebRTC streaming. With additional time, I would integrate waveform visualizers (Wavesurfer.js) and real-time multiplayer cursor presence via WebSockets.